Akanksha Muthe
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Summary

Analytical Data Science postgraduate with hands-on experience in machine learning, statistical modelling, forecasting, and large-scale data analysis. Skilled in Python, SQL, R, and data manipulation using Pandas and NumPy. Experienced in building predictive models, evaluating model performance, and translating business problems into actionable insights through projects with UK organisations including Playdale Playgrounds and Co-op UK.

Education

Lancaster University, UK
PGDip in Data Science | Sept 2024 – Oct 2025
Specialisation: Business Intelligence
Relevant Coursework: Statistical Learning, Forecasting, Intelligent Data Analysis, Data Visualisation

Pune Institute of Computer Technology, India
B.E. Information Technology | Jul 2019 – May 2023

Technical Skills

Languages: Python, SQL, R, C++, HTML/CSS
Machine Learning & Statistics: Logistic Regression, Decision Trees, Random Forests, SVM, Time Series Forecasting, PCA, Feature Engineering, Classification Models, Model Evaluation, Hypothesis Testing
Python Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, TensorFlow, Keras
Tools: Excel, Jupyter, RStudio, Github
Other: Linux, NoSQL, Cloud Concepts

Experience

Data Science Placement – Playdale Playgrounds Ltd (UK)
Jun 2025 – Sept 2025
Developed Python-based data processing workflows to analyse ~72k quote-to-order records and identify factors influencing sales conversion.
Cleaned, merged, and standardised large datasets and created reproducible analytical workflows to support business decision-making.
Performed data cleaning, preprocessing, and feature standardisation using Pandas and NumPy.
Automated analytical workflows to support reproducible business reporting and decision-making.
Delivered insights supporting strategic sales decisions and revenue planning.

Data Scientist – Co-op (Contract)
Nov 2024 – Feb 2025
Project-based
Built and evaluated machine learning and forecasting models including LightGBM, Random Forest, and SARIMA using Python.
Performed feature engineering, model training, and performance evaluation using RMSE and MAE metrics.
Compared predictive modelling approaches to improve store-level demand forecasting accuracy.

Web Developer Intern – Exposys Data Labs (Remote)
Feb 2022– Apr 2022
Developed responsive web interfaces using HTML, CSS, JavaScript.
Improved front-end performance and collaborated across cross-functional teams.

Projects

Customer Churn Analysis & Modelling (R)
Built and evaluated classification models including Logistic Regression, KNN, and Random Forest to predict customer churn, achieving 85% accuracy and identifying key behavioural drivers associated with attrition.

Emotion Detection on Social Media (CNN-BiLSTM, Word2Vec)
Built a deep learning NLP model for multi-class emotion classification.
Published findings in IJRASET, demonstrating real-world AI application.

Time Series Forecasting (R, Python)
Built and compared ETS, ARIMA, regression, and neural network models on weekly data.
Identified trend-seasonality regression as the best model using SMAPE and MASE metrics.

Climate Data Processing & Visual Analytics (Python)
Cleaned and transformed unstructured climate datasets using Pandas.
Created visualisations and exploratory analysis to identify long-term climate patterns.